Hierarchical Attention Feature Fusion-Based Network for Land Cover Change Detection With Homogeneous and Heterogeneous Remote Sensing Images

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***Abstract*— Deep learning techniques have become popular in land cover change detection (LCCD) with remote sensing images (RSIs). However, many existing networks mostly concentrate on learning deep features but without considering the effect of different features' attention and fusion strategy on detection performance. In this article, a novel hierarchical attention feature fusion (HAFF)-based network for LCCD with RSIs is proposed. In the proposed HAFF-based network, novel multiscale convolution fusion filters (MCFFs) explore the global semantic feature of the interested targets from multiperspective ways. To achieve that objective, the proposed MCFFs are composed by a well-known position attention module (PAM) and a novel multiperspectives feature filter block (MFFB) with different kernel sizes. In addition, a compound loss function was proposed for balancing the impact from the features at different levels in terms of backpropagation error. Experiments conducted on six pairs of real RSIs, including three pairs of homogeneous images and three pairs of heterogeneous images, confirmed the superiority of the proposed HAFF network over other cognate methods. Moreover, the ablation experiments further confirmed the feasibility and superiority of the proposed MCFFs, whereas quantitative observations indicated that competitive improvements are achieved by the proposed MCFFs in terms of all the evaluation indicators. The code for the proposed approach will be available at https://github.com/ImgSciGroup/HAFF.**

***Index Terms*— Change detection, deep learning, remote sensing images (RSIs), UNet.**

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I. Introduction

LAND cover change detection (LCCD) is the process of quantitatively analyzing Earth's surface change with bitemporal remote sensing images (RSIs) [1]. LCCD is defined by achieving change detection by comparing pre-event and post-event images, and LCCD with RSI has been known for its advantages of "large area coverage" and "rapid acquisition" for determining land cover change information. Therefore, LCCD with RSIs has been applied successfully in land use analysis [2], [3], landslide disaster [4], [5], flood detection [6], and forest deformation [7], among others, and has become popular in recent decades [8], [9], [10].

Various change detection methods with RSIs have been proposed in the past decades, and they can be divided into two major groups, including traditional methods (non-deep learning-based methods) [9], [11], [12] and deep learning methods [13], [14], [15], the details about these studies are presented below.

Traditional change detection methods with RSIs generally concentrate on improving change detection performance by exploring spatial-contextual features around a central pixel [10], such as local spectral similarity measurement [16], [17] and local texture extraction [18], [19]. In addition, the mathematical model is popular for describing neighboring information, such as multiresolution level set (MLS) [20], change vector analysis fuzzy C-means (CVA-FCM) [21], and Markov random field-based methods [22]. These methods are suitable for LCCD with bitemporal homogeneous RSIs. However, bitemporal homogenous RSIs may be unavailable in many practical application scenarios, such as landslide or flood disasters that occur at night, and then the bitemporal

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optical images were unavailable timely for locating land cover change areas and carrying out rescue. To achieve LCCD with multimodal images, many methods have been proposed for LCCD with heterogeneous RSIs, such as Markov random field-based approaches [23], extended multiattribute profiles approaches [24], adaptive graph and structure cycle consistency (AGSCC) approaches [25], structure regression fusion [26], and structure transformation [27]. The heterogeneous images usually contain optical satellite and SAR images, and some very promising solutions have been introduced that can improve the observation efficiency of future spaceborne imaging radars [28], [29]. However, different feature representation of bitemporal heterogeneous RSIs leads to the problem that they cannot be compared directly for LCCD. LCCD with heterogeneous RSIs aims at exploring shared features from multimodule images and measuring the change magnitude to achieve the LCCD task, which is an important supplement for LCCD with homogeneous RSIs [9]. Therefore, proposing an approach that is robust for both homogeneous and heterogeneous RISs for LCCD is preferred in practical applications.

In addition to traditional methods (non-deep learning methods), deep learning methods are currently popular in exploring deep features that are insensitive to the external factor's change of bitemporal RSIs, including illumination caused by sun-height, moisture caused by weather, and phenological difference caused by seasons. For such conditions, deep learning often obtains satisfactory detection maps with RSIs [30]. Therefore, deep learning-based LCCD has been widely used for LCCD with RSIs [31]. One classical neural network that is widely used in the community of LCCD with RSIs is the U-Net architecture [32], [33]. U-Net was first used for biomedical image segmentation tasks [34], and its architecture stems from the so-called "fully convolutional network," which can achieve fast and precise segmentation [35], [36]. For change detection tasks, U-Net allows the network to simultaneously explore global and contextual information to obtain better detection performance. Owing to the traits of U-Net, various U-net-based methods, such as fully connected Siam connection (FC-Siam-conc) [37], dense connected Siamese network-based U-Net [38], [39], [40], modified U-Net for change detection with hyperspectral images [41], and residual U-Net for building change detection with high-resolution RSIs [42], [43], [44], have been promoted in practical applications. All LCCD performances in the preceding U-Net studies have provided evidence to support the backbone of U-Net's success in terms of its simplicity and superiority in practical applications. Wu et al. [45] proposed a fully convolutional change detection framework with a generative adversarial network with the U-net backbone to unify unsupervised, weakly supervised, regional supervised, and fully supervised change detection tasks, significantly indicating that research on UNet is still ongoing.

Apart from studies on the classical U-Net in the community of LCCD with RSIs, attention mechanisms were attractive and embedded in many deep learning networks to acquire better change detection. For example, multiscale attention [46], [47], [48], high-frequency attention [49], deeply supervised attention metric-based [50], and hierarchical attention networks [51] have promoted LCCD with RSIs. In these approaches, convolutional block attention modules, including channel attention and spatial attention, are suggested to embed in the backbone of a neural network [52]. Owing to the advantages of the attention mechanism, networks with attention devoted considerable focus to the important components of the input dataset, even though these components are small. Therefore, deep learning change detection methods with attention mechanisms can achieve better detection maps than conventional methods [53].

Apart from deep and effective feature learning approaches, feature fusion is another strategy to enhance the detection performance with RSIs. The reason is that features from different perspectives are useful for improving the distinguishability of a target. Feng et al. [54] developed an inter-scale feature fusion network for LCCD with bitemporal RSIs. Luo et al. [55] promoted a multiscale diff-changed feature fusion network for LCCD with hyperspectral RSIs and acquired satisfactory detection maps. Jiang et al. [56] proposed an LCCD approach with optical and SAR images via deep homogeneous feature fusion. Zhang et al. [57] fused the deep supervised features for LCCD with very high-resolution RSIs. Liu et al. [58] combined prior extraction and contextual fusion by learning prior-aware transformers for building change detection with RSIs. From the preceding review, various applications indicated that feature fusion is a feasible and effective way for improving detection performance. However, the performance of these approaches usually depends on datasets, and most of them have been demonstrated to be impractical [59], [60].

The preceding review indicates that although deep learning technique has been proposed and applied successfully in LCCD with RSIs, particularly using the backbone of U-Net with attention mechanism and feature fusion strategy, it still has some limitations and remains an improvement space for LCCD, as presented below.

1) Exploring deep features with fixed convolution kernel cannot precisely cover the changing area with different shapes and sizes. Intuitively, change and unchanged areas in an image scene may have different shapes and sizes, and a neural network with a fixed-size kernel convolution cannot learn the feature precisely. Although multiscale convolution has been applied to reduce the negative effect of the various shapes and sizes of different targets, the global semantic feature of an interesting target in an image scene is often disregarded in the existing methods. How to learn the features from different convolution layers precisely remains a challenge for LCCD with RSIs.
2) Directly using the learned features from different levels cannot achieve a satisfactory detection performance. The reason is that using the same loss function for learning the feature from different levels of convolution layers without considering the feature's individuality from different layers plays different roles in recognizing

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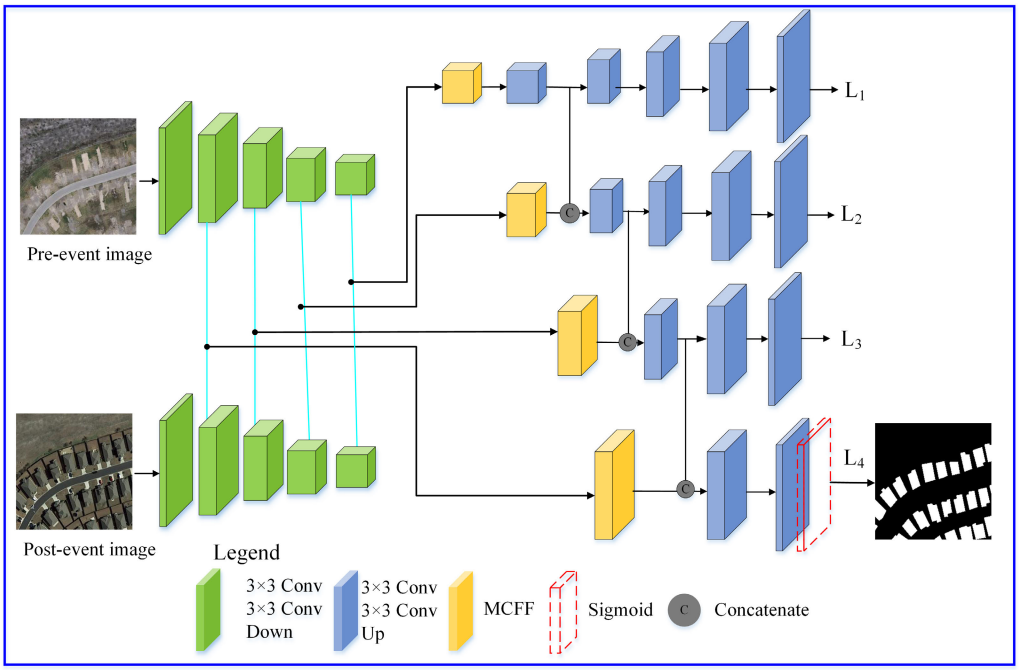


Fig. 1. Flowchart of the proposed HAFF-based network for LCCD with bitemporal RSIs.

a target. How to fuse the learned feature from different layers remains a challenge while using U-Net for improving LCCD performance with RSIs.

The motivation of this article lies in proposing a novel hierarchical attention feature fusion (HAFF) approach and improves change detection performance. The major contributions of the article are summarized as follows.

1) A novel HAFF-based network is proposed for LCCD with RSIs. In particular, it is demonstrated that the HAFF approach is feasible and robust for homogeneous and heterogeneous RSIs in LCCD. The comparative results with several cognate methods indicate the superiority of the proposed HAFF-based network.
2) A multiscale convolution fusion filter (MCFFs) is constructed with different kernel sizes aiming at covering various targets. In the proposed MCFFs, a novel multiperspective feature filter block (MFFB) is also proposed to determine the global semantic feature of the interested targets. In the experiments, ablation investigations clearly demonstrated that the proposed MCFFs are effective in improving detection performance.
3) A novel multilevel composite loss function [61] based on dice loss was developed to reduce the backpropagation error from different levels. Dice loss was particularly adopted for balancing the effect of background information. Moreover, different attention feature maps are eventually fused to obtain a change probability map with cross-loss function [62].

The remainder of this article is organized as follows. Section II presents the details of the proposed approach. Section III provides the experiments. Finally, Section IV presents the conclusion.

II. Methodology

As shown in Fig. 1, the proposed HAFF-based network is an end-to-end network. Different from the backbone of U-Net, an MCFF module was developed and embedded in the proposed HAFF-based network before the decoder. The objective is to compel the neural network to focus on determining the global semantic feature of the interested targets. In addition, dice loss was used in the decoder for upscaling different features.

Details of the proposed HAFF-based network are presented in terms of neural architecture. The network contains an encoder, MCFF module, and a decoder. Inputs of the proposed network are the bitemporal RSIs, and the output is a binary change detection map. Each layer in the encoding progress has two 3×3 convolutions and a 2×2 max pooling, and each layer in the decoding progress has two 3×3 deconvolutions. In addition, a novel MCFF was embedded at the beginning of decoding. Details of the proposed MCFF are presented as follows.

A. MCFFs Module

The proposed MCFFs module was developed to focus on learning the multiscale information of interesting targets. As shown in Fig. 2, a well-known module called position attention module (PAM), which has been widely used in the field of computer vision [63], [64], [65], is used in the proposed MCFFs module to explore a richer contextual relationship over an input local feature (F_{in}). Thereafter, the novel multi-MFFB with different kernel sizes are parallelized to extract different features. Finally, extracted features from different perspectives are concatenated and reshaped to be an output feature (F_{out}).

In addition, the structure of PAM is given in Fig. 3, showing that a given feature $F_{\text{in}} \in R^{H \times W \times C}$ is fed into three

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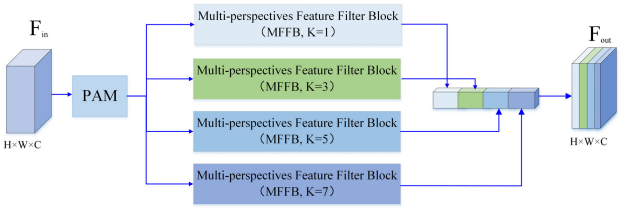


Fig. 2. Flowchart of the proposed multiscale convolution feature fusion module.

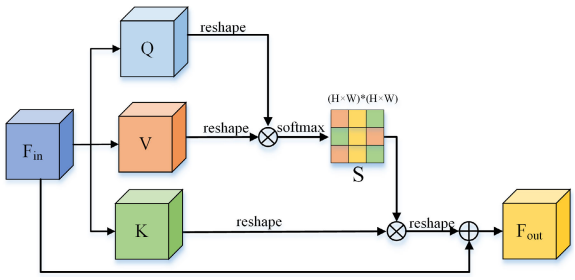


Fig. 3. Structure of the PAM.

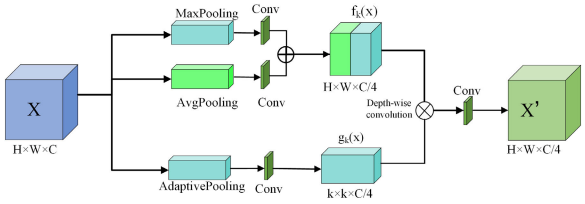


Fig. 4. Proposed structure of the multi-MFFB.

parallelized convolution layers with 3 × 3 kernel size. Then, the two learned features (i.e., Q and V) are reshaped to $R^{C\times N}$, where $N = H \times W$ is the number of pixels. Finally, a matrix multiplication between the transformed Q and V is applied to calculate the spatial position attention map (S)

$$S_{i,j} = \frac{\exp(Q_i \cdot V_j)}{\sum_{i=1}^{i=N} \exp(Q_i \cdot V_j)} \tag{1}$$

where $S_{i,j}$ measures the relationship between the ith position at feature map Q and the jth position in feature V. Meanwhile, F_{in} is also fed into a convolution layer to obtain a new feature K, where $K \in R^{H\times W\times C}$ is reshaped to $R^{N\times N}$. Finally, the output feature is computed as follows:

$$F_j^{\text{out}} = \alpha \sum_{i=1}^{N} (S_{ij} K_i) + F_j^{\text{in}} \tag{2}$$

where α is a scale parameter. The initial value of α is zero, but it is gradually learned and updated. Equation (2) shows that a similar feature representation will present a similar value of $S_{i,j}$ and will generate a similar output feature. Owing to the weight summarized in all positions in an image scene, the global contextual effect can be aggregated and generated into a spatial attention map S.

After an input feature is processed with PAM, the feature of interesting targets can be enhanced. However, a target in bitemporal RSIs is often varied by the change of illumination, season, and phenology. That phenomenon may cause a pseudo-change in a detection map. Therefore, the MFFB was proposed

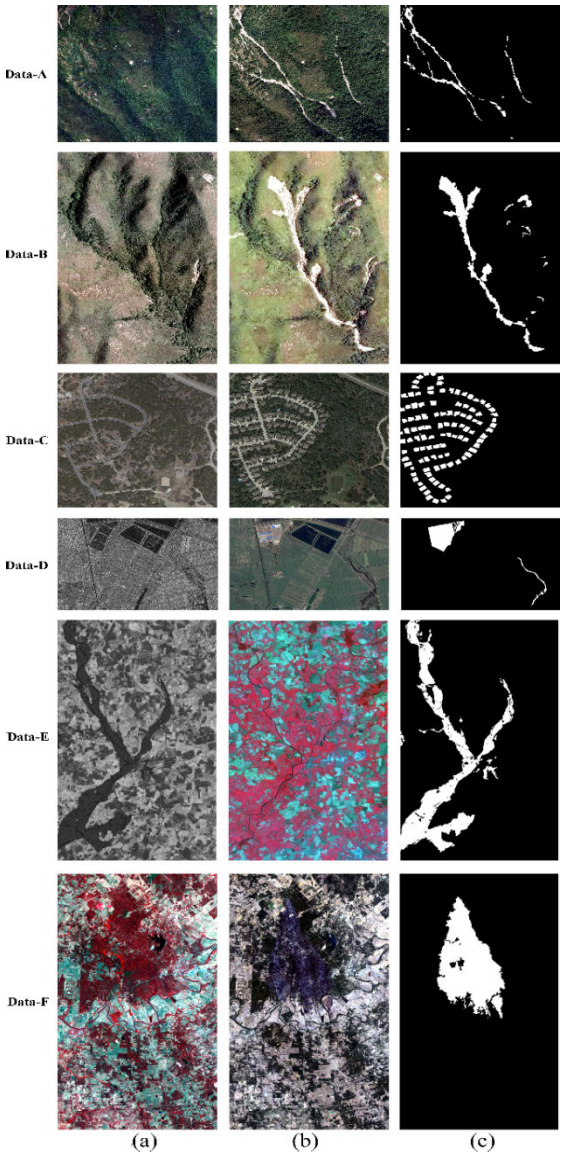


Fig. 5. Datasets A–F (a) pre-event image, (b) post-event image, and (c) ground reference map.

to further improve the distinguishability of change area. As shown in Fig. 4, the input feature (X) in MFFB is fed into two convolution layers with Max-pooling and Avg-Pooling to learn the feature from different perspectives of a target. Then, the learned features are processed by convolution layers with a 3 × 3 kernel size. Finally, the multiperspective features are concatenated and reshaped into a new feature $f_k(x)$.

In the MFFB, the input feature (X) is fed into a convolution layer with adaptive pooling to generate a reduced feature map $g_k(x) \in R^{H\times W\times C'}$, where H and W are the height and width, respectively, of the obtained feature, C' is the number of channels for the input feature, and k is the kernel size of the filter. After the convolutional operation, the obtained features that are extracted from different multiperspectives using different pooling approaches are fused by a depth-wise convolution

$$X' = f_k(x) \otimes g_k(x) \tag{3}$$

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TABLE I
SUMMARIZED DESCRIPTION OF THE DATASETS

Data set	Dates	Type	Location	Event	Resolution (m/pixel)	Size (pixel)
Data-A	Pre-event Image, December 2007	Zeiss RMK TOP 15	Hong Kong, China	Landslide	0.5 m/pixel	1024 × 2024
	Post-event Image, November 2014	Zeiss RMK TOP 15	Hong Kong, China	Landslide	0.5 m/pixel	1024 × 2024
Data-B	Pre-event Image, April 2007	Zeiss RMK Top-1	Hong Kong, China	Landslide	0.5 m/pixel	593 × 921
	Post-event Image, July 2014	Zeiss RMK Top-1	Hong Kong, China	Landslide	0.5 m/pixel	593 × 921
Data-C	Pre-event Image, 2002	Google Earth	Texas, USA	Building change	0.5 m/pixel	1024 × 2024
	Post-event Image, 2018	Google Earth	Texas, USA	Building change	0.5 m/pixel	1024 × 2024
Data-D	Pre-event Image, June 2008	Radarsat-2	Shuguang Village, China	Building construction	8.0 m/pixel	921 × 593
	Post-event Image, September 2012	Google Earth	Shuguang Village, China	Building construction	8.0 m/pixel	921 × 593
Data-E	Pre-event Image, 1999	SPOT	Gloucester, England	Flooding	25.0 m/pixel	554 × 990
	Post-event Image, 2000	NDVI	Gloucester, England	Flooding	25.0 m/pixel	554 × 990
Data-F	Pre-event Image, August 2011	Landsat-5 TM	Texas, USA	Forest fire	30.0 m/pixel	808 × 1534
	Post-event Image, June 2013	Landsat-5 TM	Texas, USA	Forest fire	30.0 m/pixel	808 × 1534

TABLE II
COMPARISON BETWEEN THE PROPOSED HAFF AND THE EIGHT STATE-OF-THE-ART RELATED METHODS FOR DATA-A TO -C, Ka ∈ [0, 1], AND OTHERS ARE PRESENTED IN PERCENTAGE (%). THE BEST RESULTS ARE GIVEN IN BOLD

Data sets	Methods	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	Precision	Recall	F1
Data-A	MLS [20]	93.96	0.46	**93.03**	5.98	7.96	6.04	31.68	93.79	62.74	32.57	92.04	48.12
Data-B		89.73	0.42	89.67	10.26	10.40	10.27	29.77	89.26	59.52	30.84	89.60	45.88
Data-C		69.00	-0.03	47.48	25.05	79.98	31.00	6.53	68.32	37.42	8.84	20.02	12.26
Data-A	CVA-FCM [21]	86.54	0.26	89.79	13.67	**6.74**	13.46	17.41	86.14	51.77	17.63	**93.26**	29.65
Data-B		80.45	0.25	85.75	20.13	8.38	189.55	18.54	79.53	49.04	18.86	91.62	31.28
Data-C		62.85	-0.01	48.84	33.28	69.03	37.15	8.28	61.56	34.92	10.15	30.97	15.29
Data-A	AGSCC [25]	97.80	0.63	81.34	1.13	36.20	2.20	46.88	97.76	72.32	63.87	63.80	63.84
Data-B		97.85	0.75	85.78	0.86	27.59	2.15	62.02	97.77	79.90	81.21	72.41	76.56
Data-C		85.58	0.07	52.72	5.34	89.22	14.42	7.49	85.41	46.45	19.69	10.78	13.94
Data-A	Siam-U-net-Conc [37]	98.97	0.80	84.62	0.11	30.64	1.04	67.10	98.94	83.02	95.37	69.36	80.31
Data-B		98.70	0.85	90.95	0.46	17.64	1.30	75.51	98.65	87.08	90.08	82.36	86.05
Data-C		98.10	0.90	92.46	0.35	14.73	1.90	82.90	97.90	90.40	96.75	85.27	90.65
Data-A	Siam-U-net-Diff [37]	98.45	0.66	75.37	0.05	49.20	1.55	49.98	98.43	74.21	**96.90**	50.80	66.65
Data-B		97.89	0.73	80.26	**0.21**	39.26	2.11	58.31	97.83	78.07	**93.58**	60.74	73.67
Data-C		94.95	0.67	76.81	**0.04**	46.34	5.06	53.47	94.63	74.05	**99.33**	53.66	69.68
Data-A	SNU-Net [40]	99.03	0.81	85.72	0.11	28.45	0.97	69.09	99.00	84.05	95.25	71.55	81.72
Data-B		98.66	0.86	93.97	0.83	11.23	1.34	76.30	98.60	87.45	84.46	88.77	86.56
Data-C		98.09	0.90	92.54	0.37	14.55	1.91	82.91	97.90	90.41	96.55	85.45	90.66
Data-A	DSAM-Net [50]	98.00	0.51	67.73	**0.04**	64.49	2.00	35.08	97.98	66.53	96.69	35.51	51.94
Data-B		98.31	0.81	89.28	0.72	20.71	1.69	69.46	98.24	83.85	84.86	79.29	81.98
Data-C		97.33	0.85	88.11	0.12	23.66	2.67	75.60	97.09	86.35	98.73	76.34	86.11
Data-A	PATM-Net [58]	98.82	0.78	86.38	0.37	26.87	1.18	65.39	98.80	82.09	86.07	73.13	79.08
Data-B		98.37	0.82	91.28	0.86	16.58	1.63	71.34	98.30	84.82	83.13	83.42	83.27
Data-C		97.90	0.88	91.50	0.33	16.67	2.10	81.10	97.69	89.40	96.80	83.33	89.57
Data-A	Proposed HAFF	**99.10**	**0.83**	87.73	0.16	24.38	**0.90**	**71.98**	**99.08**	**85.53**	93.73	75.62	**83.71**
Data-B		**98.80**	**0.88**	**95.55**	0.85	**8.06**	**1.20**	**78.85**	**98.75**	**88.80**	84.71	**91.94**	**88.18**
Data-C		**98.12**	**0.90**	**94.15**	0.78	**10.91**	**1.88**	**83.68**	**97.92**	**90.80**	93.23	**89.09**	**91.12**

where X' is the filtered feature and $\otimes$ denotes a depth-wise convolution.

B. Proposed Compound Loss Function

The aforementioned network is used as a basis to propose a compound loss function (CLF) for training the neural network. CLF contains four parts and each part has a different weight to balance the effect of different level features as shown below

$$\mathrm{CLF} = \lambda_1 L_1 + \lambda_2 L_2 + \lambda_3 L_3 + \lambda_4 L_4 \tag{4}$$

$$L_1 = L_2 = L_3 = 1 - \frac{2\sum_{i=1}^{i=N} y_i p_i}{\sum_{i=1}^{i=N} y_i + \sum_{i=1}^{i=N} p_i} \tag{5}$$

$$L_4 = \frac{1}{N}\sum [y_i \log p_i + (1 - y_i)\log(1 - p_i)] \tag{6}$$

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TABLE III

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Data sets	Methods	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	Precision	Recall	F1
Data-D	MLS [20]	85.33	0.21	72.21	13.34	42.24	14.67	15.32	84.93	50.13	17.26	57.76	26.57
Data-E		79.77	0.35	75.49	18.89	30.14	20.23	29.17	77.93	53.55	33.37	69.86	45.17
Data-F		79.13	0.26	69.60	18.30	42.51	20.87	22.66	77.77	50.21	27.22	57.49	36.95
Data-D	CVA-FCM [21]	74.98	0.13	70.57	24.57	34.29	25.02	10.77	74.20	42.49	11.41	65.71	19.45
Data-E		64.27	0.20	71.19	37.90	19.72	35.73	21.14	60.49	40.81	22.29	80.28	34.90
Data-F		67.41	0.10	59.73	30.51	50.02	32.59	14.03	65.58	39.80	16.32	49.98	24.61
Data-D	AGSCC [25]	98.26	0.77	83.83	0.28	32.06	1.74	64.19	98.20	81.20	92.09	67.94	78.19
Data-E		95.48	0.79	90.79	3.05	15.37	4.52	69.07	94.97	82.02	78.98	84.63	81.71
Data-F		97.60	0.88	94.72	1.62	8.94	2.40	80.13	97.34	88.74	86.98	91.06	88.97
Data-D	Siam-U-net-Conc [37]	97.09	0.54	69.25	**0.09**	61.40	2.91	37.87	97.04	67.45	**95.28**	38.60	54.94
Data-E		97.78	0.90	97.16	2.02	3.65	2.22	83.83	97.50	90.66	86.58	96.35	91.20
Data-F		98.06	0.89	91.73	0.23	16.31	1.94	82.10	98.87	89.98	97.73	83.69	90.17
Data-D	Siam-U-net-Diff [37]	96.90	0.52	69.17	0.29	61.36	3.10	36.44	96.85	66.64	86.47	38.64	53.41
Data-E		96.05	0.81	87.74	**1.23**	23.29	3.95	70.53	95.64	83.09	**89.76**	76.71	82.72
Data-F		97.97	0.90	97.99	2.04	1.99	2.04	83.67	97.73	90.70	85.12	**98.02**	91.11
Data-D	SNU-Net [40]	98.50	0.79	88.34	0.75	22.58	1.75	67.03	98.19	82.61	83.32	77.42	80.26
Data-E		97.50	0.89	**97.84**	2.61	**1.71**	2.50	82.41	97.17	89.79	83.61	**98.29**	90.36
Data-F		98.87	0.94	94.84	**0.04**	10.28	1.13	89.41	98.75	94.08	**99.62**	89.72	94.41
Data-D	DSAM-Net [50]	97.92	0.74	83.84	0.65	31.67	2.08	60.18	97.85	79.02	83.46	68.33	75.14
Data-E		96.20	0.81	88.52	1.39	21.57	3.80	71.13	95.81	83.47	88.42	78.43	83.13
Data-F		98.58	0.92	94.17	0.23	11.43	1.42	86.88	98.43	92.66	97.86	88.57	92.98
Data-D	PATM-Net [58]	83.44	0.21	76.53	15.86	31.07	16.56	16.06	82.90	49.48	17.31	68.93	27.67
Data-E		97.58	0.89	96.89	2.21	4.01	2.42	82.53	97.26	89.89	85.48	95.99	90.43
Data-F		98.67	0.93	94.61	0.23	10.54	1.33	87.77	98.53	93.15	97.89	89.46	93.49
Data-D	Proposed HAFF	**99.17**	**0.91**	**95.52**	0.46	**8.50**	**0.83**	**83.47**	**99.13**	**91.30**	90.49	**91.50**	**90.99**
Data-E		**97.96**	**0.91**	97.81	1.99	2.39	**2.04**	**85.10**	**97.69**	**91.40**	86.91	97.61	**91.95**
Data-F		**99.17**	**0.96**	**97.80**	0.46	**3.95**	**0.83**	**92.49**	**99.08**	**95.78**	96.15	96.05	**96.10**

where $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1$. To reduce the negative effect of the imbalance between the background and the changed area, λ_1, λ_2, λ_3 are dice loss functions, the purpose of (5) lies in reducing the negative effect from the imbalance samples. Given that a binary change detection task can be considered a binary classification map, L_4 is given as a cross-entropy loss function, as shown in (6), where y_i is the label of the ith sample, $y_i = 0$ and $y_i = 1$ indicate the unchanged and changed samples, respectively, and p_i is the prediction probability for change class.

III. EXPERIMENTS

In this section, the proposed HAFF-based network was applied to real datasets to test its feasibility for LCCD with RSIs. Eight cognate methods, namely MLS [20], CVA-FCM [21], AGSCC [25], Siam-U-net-Conc [37], Siam-U-net-Diff [37], SNU-Net [40], DSAM-Net [50], and PATM-Net [58], were selected for performance comparisons. We did not compare with multilayer perceptron (MLP) and support vector machine (SVM)-based approaches because comparing with some state-of-the-arts methods is more persuasive. Details of the comparisons are given below.

A. Data Description

Six pairs of bitemporal RSIs, including three pairs of homogeneous images and three pairs of heterogeneous images, were used in experiments. As listed in Table I, these datasets were acquired with different platforms using different remote sensors, such as aerial cameras, Landsat satellites, and Spot satellite. Fig. 5 shows that the datasets are related to different change events, including landslides, buildings, floods, and wildfires. More details about the datasets can be found in [66]. Achieving the land cover change task with the aforementioned datasets involves two challenges.

1) First, homogeneous bitemporal RSIs used in LCCD may be affected by external imaging condition when the images are captured. Although bitemporal RSI images cover the same geographical area and share the same features on the ground, the quality of an image may be affected by different seasons, illumination, and sun height, as shown in Fig. 5(a)–(c). External differences may cause pseudo-changes in a detection map, detecting real change, while inhibiting the pseudo-change in a detection map remains a challenge.

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TABLE IV
DETAILED DESCRIPTION OF EACH EVALUATION INDICATOR USED IN QUALITATIVE COMPARISON

Evaluation Indicators	Formula	Explanation
Overall Accuracy (OA)	$OA = \frac{TP+TN}{TP+TN+FP+FN}$	Ratio between the correctly detected pixels in the result and the total number of the ground truth map
Kappa Coefficient (Ka)	$P_0 = \frac{TP+TN}{TP+TN+FP+FN}$ $P_e = \frac{(TP+FP)\times(TP+FN)+(FN+TN)\times(FP+TN)}{(TP+TN+FP+FN)^2}$ $Ka = \frac{P_0-P_e}{1-P_e}$	Ka reflects the reliability of the detection map by measuring the inter-rater reliability for the changed and unchanged items.
Average Accuracy (AA)	AA = $(\frac{TP}{TP+FP} + \frac{TN}{TN+FN})/2$	Average accuracy for the corrected change detection accuracy and unchanged detection accuracy.
False Alarm (FA)	$FA = \frac{FP}{TN+FP}$	Ratio between the number of pixels that are detected as changed pixels but are unchanged in the ground truth map and the number of pixels that are unchanged in the ground truth map.
Missing Alarm (MA)	$MA = \frac{FN}{TP+FN}$	Ratio between the number of pixels that are identified as unchanged but are changed in the ground truth map and the number of pixels that are changed in the ground truth map.
Total Error (TE)	$TE = \frac{FP+FN}{TP+TN+FP+FN}$	Ratio betwen the number of incorrectly detected pixels in the result and the total number of the ground truth map.
F1-Score	$F1 = 2 \times \frac{precision \times recall}{precision+recall}$	F1 score is an indicator used to measure the accuracy of the dichotomous model in statistics. It considers the accuracy and recall of classification models. The F1 score can be regarded as a harmonic average of model accuracy and recall.

2) Second, heterogeneous bitemporal RSIs cannot compare directly to measure change magnitude, but they are important for LCCD when homogeneous RSIs are unavailable. Heterogeneous images acquired with different sensors, such as optical and SAR sensors, and RSIs acquired with different remote sensors express the ground target from different perspectives. For example, the appearance of ground targets in an optical image depends on the shape, size, and external illumination light condition. By contrast, the appearance of a target in an SAR image depends on the roughness of the target's surface and the material's physical characteristics. Therefore, making bitemporal heterogeneous RSIs to be comparable to LCCD remains a challenge.

In the following comparison experiments, we selected the aforementioned datasets to verify the feasibility, robustness, and superiority of our proposed approach for achieving LCCD with homogeneous and heterogeneous RSIs.

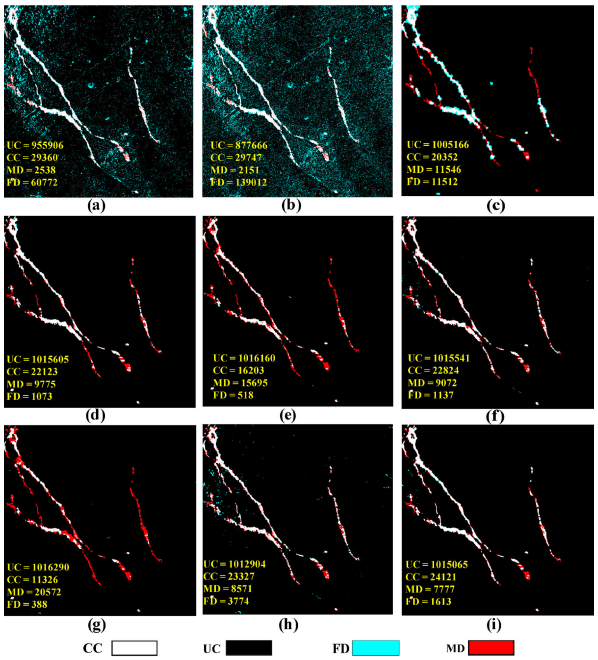


Fig. 6. Detection maps acquired with different methods for Data-A (a) MLS [20], (b) CVA-FCM [21], (c) AGSCC [25], (d) Siam-U-net-Conc [37], (e) Siam-U-net-Diff [37], (f) SNU-Net [40], (g) DSAM-Net [50], (h) PATM-Net [58], and (i) proposed HAFF.

B. Experiment Setting

Eight cognate and state-of-the-art methods were selected for comparison to verify the superiority of the proposed approach. The characteristics and parameters of each approach are as follows.

1) *MLS [20]:* This method is carried out by means of a level set method which iteratively seeks to find a global optimal contour which splits the image into two mutually exclusive regions associated with changed and unchanged classes, respectively.

2) *CVA-FCM [21]:* This method is based on the proposed sequential spectral change vector analysis, which exploits an iterative hierarchical scheme that iteratively discovers and identifies a subset of changes.

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TABLE V

QUANTITATIVE COMPARISON IN TERMS OF THE AREAS UNDER THE RECEIVER OPERATING CHARACTERISTICS CURVE (AUR) AND THE AREAS UNDER THE PRECISION-RECALL CURVE (AUP) FOR CMIS GENERATED BY DIFFERENT METHODS ON DATASET-A TO -F

Methods	Data-A		Data-B		Data-C		Data-D		Data-E		Data-F		Average	
	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP
MLS	0.803	0.523	0.734	0.529	0.494	0.340	0.690	0.438	0.579	0.547	0.638	0.427	0.656	0.467
CVA-FCM	0.964	0.773	0.938	0.666	0.482	0.100	0.760	0.186	0.731	0.207	0.658	0.182	0.756	0.352
AGSCC	0.965	0.696	0.961	0.806	0.803	0.273	0.961	0.791	0.941	0.763	0.970	0.824	0.933	0.692
Siam-U-net-Conc	0.963	0.926	**0.998**	**0.958**	0.959	0.915	0.943	0.936	0.996	**0.980**	0.993	0.985	0.975	0.950
Siam-U-net-Diff	0.850	0.778	0.995	0.925	0.984	**0.975**	0.891	0.879	0.994	0.953	**0.999**	**0.993**	0.952	0.917
SNU-Net	0.984	0.918	0.923	0.876	0.943	0.944	0.852	0.859	0.991	0.962	0.924	0.924	0.936	0.914
DSAM-Net	0.894	0.830	0.875	0.807	0.836	0.843	0.962	0.927	0.979	0.943	0.992	0.961	0.923	0.885
PATM-Net	0.981	0.904	0.971	0.872	0.977	0.965	0.956	0.952	0.989	0.975	0.975	0.956	0.975	0.937
Proposed HAFF	**0.995**	**0.940**	0.995	0.952	**0.989**	0.964	**0.992**	**0.953**	**0.997**	0.971	0.988	0.982	**0.993**	**0.960**

TABLE VI

QUANTITATIVE COMPARISON IN TERMS OF PRECISION (PRE) AND RECALL (REC) GENERATED BY DIFFERENT METHODS ON DATASET-A TO -F

Methods	Data-A		Data-B		Data-C		Data-D		Data-E		Data-F		Average	
	PRE	REC	PRE	REC	PRE	REC	PRE	REC	PRE	REC	PRE	REC	PRE	REC
MLS	32.57	92.04	30.84	89.60	8.84	20.02	17.26	57.76	33.37	69.86	27.22	57.49	25.02	64.46
CVA-FCM	17.63	**93.26**	18.86	91.62	10.15	30.97	11.41	65.71	22.29	80.28	16.32	49.98	16.11	68.64
AGSCC	63.87	63.80	81.21	72.41	19.69	10.78	92.09	67.94	78.98	84.63	86.98	91.06	70.47	65.10
Siam-U-net-Conc	95.37	69.36	90.08	82.36	96.75	85.27	**95.28**	38.60	86.58	96.35	97.73	83.69	93.63	75.94
Siam-U-net-Diff	**96.90**	50.80	**93.58**	60.74	**99.33**	53.66	86.47	38.64	**89.76**	76.71	85.12	**98.02**	**91.86**	63.10
SNU-Net	95.25	71.55	84.46	88.77	96.55	85.45	83.32	77.42	83.61	**98.29**	**99.62**	89.72	90.47	85.20
DSAM-Net	96.69	35.51	84.86	79.29	98.73	76.34	83.46	68.33	88.42	78.43	97.86	88.57	91.67	71.08
PATM-Net	86.07	73.13	83.13	83.42	96.80	83.33	17.31	68.93	85.48	95.99	97.89	89.46	77.78	82.38
Proposed HAFF	93.73	75.62	84.71	**91.94**	93.23	**89.09**	90.49	**91.50**	86.91	97.61	96.15	96.05	90.87	**90.30**

3) *AGSCC [25]:* This method is means A structure cycle consistency-based image regression method, which consists of two components: exploration of structure representation and structure-based regression. Then, change extraction can be achieved by directly comparing the transformed and original images.
4) *Siam-U-Net-Conc [37]:* This network is an improved version of the classical U-Net. In contrast to the classical U-Net, the selected Siam-U-Net-Conc network is an extension of full connection convolutions. Moreover, the difference is that FC-Siam-Conc skips two connections in the decoding process.
5) *Siam-U-Net-Diff [37]:* In contrast to Siam-U-Net-Conc [37], the selected Siam-U-Net-Diff calculates the change magnitude via image by image difference in the learning process.
6) *SNU-Net [40]:* The selected SNU-Net considers the ignorance of the important shallow feature of the deep learning of change detection, and alleviates the loss of localization information through compact information transmission between encoder and decoder. SNU-Net can refine the representative features of different semantic levels and obtain satisfactory detection maps with homogeneous RSIs.
7) *DSAM-Net [50]:* In the selected DSAM-Net, a deep supervised attention module, which contains deeply supervised and convolutional block attention modules, was proposed for acquiring change detection maps with real bitemporal datasets.
8) *PATM-Net [58]:* The selected PATM-Net concentrates on exploring prior and deep features from bitemporal RSIs for change detection tasks with buildings. Moreover, it combines prior extraction and contextual fusion by learning a prior-aware transformer for obtaining a better detection map.

To guarantee comparative fairness, parameters for all the deep learning methods are the same, i.e., learning rate $= 0.001$; batch size $= 64 \times 64$; and the ratio between training and test samples is 8:2. The detailed description of the evaluation metrics is summarized in Table IV. The details about other evaluation metrics, including UCIoU, CIoU, and mIoU, can be found in [67].

C. Experimental Results

On the basis of the parameter set discussed above, the detection map achieved by the selected methods and our proposed approach are compared in detail below.

1) *Comparison Based on Bitemporal Homogenous RSIs (Data-A to -C):* The quantitative comparisons are

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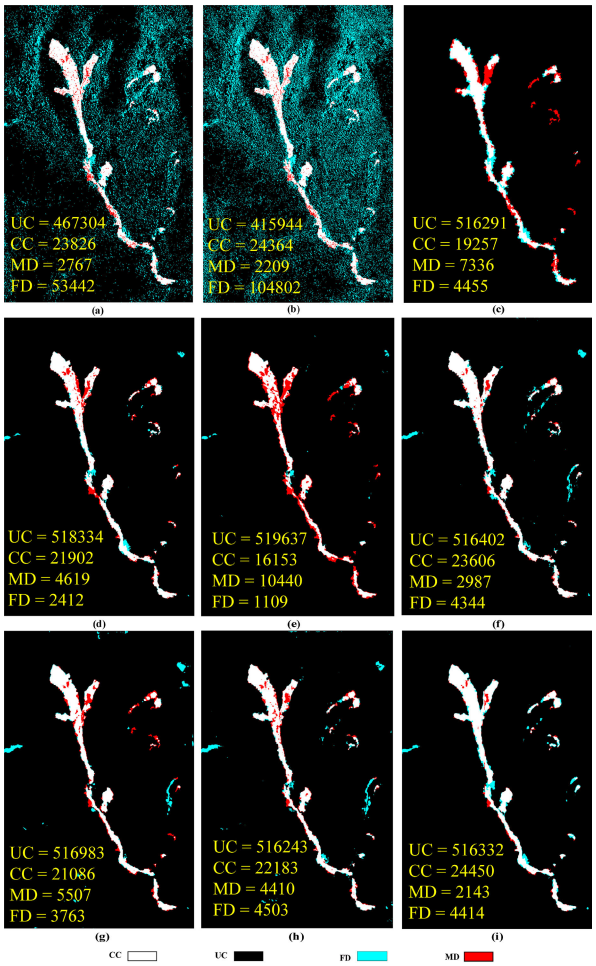


Fig. 7. Detection maps acquired with different methods for Data-B (a) MLS [20], (b) CVA-FCM [21], (c) AGSCC [25], (d) Siam-U-net-Conc [37], (e) Siam-U-net-Diff [37], (f) SNU-Net [40], (g) DSAM-Net [50], (h) PATM-Net [58], and (i) proposed HAFF.

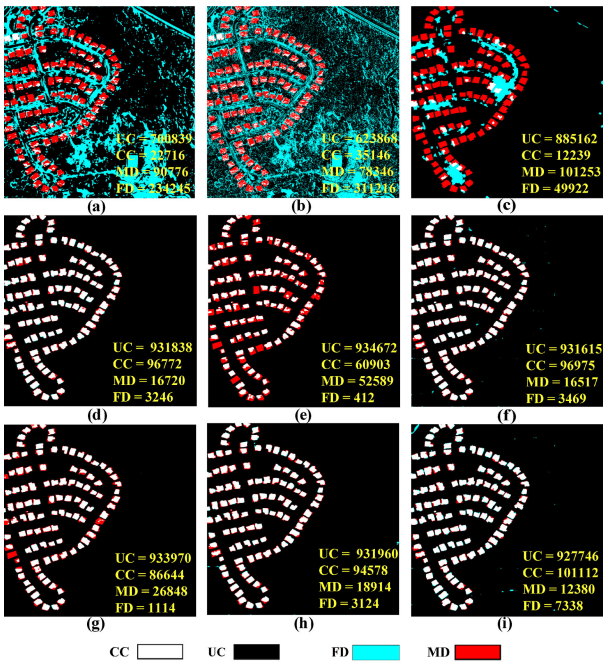


Fig. 8. Detection maps acquired with different methods for Data-C (a) MLS [20], (b) CVA-FCM [21], (c) AGSCC [25], (d) Siam-U-net-Conc [37], (e) Siam-U-net-Diff [37], (f) SNU-Net [40], (g) DSAM-Net [50], (h) PATM-Net [58], and (i) proposed HAFF.

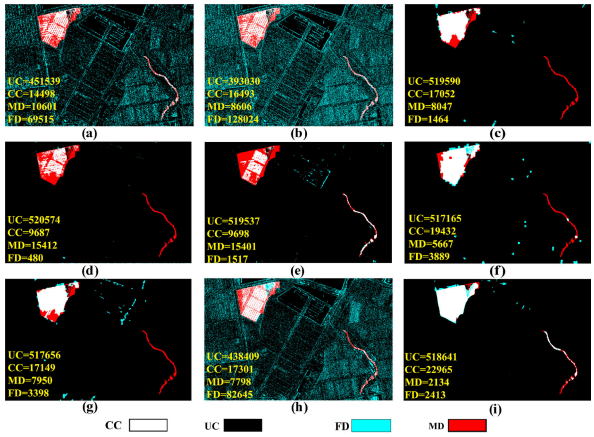


Fig. 9. Detection maps acquired with different methods for Data-D (a) MLS [20], (b) CVA-FCM [21], (c) AGSCC [25], (d) Siam-U-net-Conc [37], (e) Siam-U-net-Diff [37], (f) SNU-Net [40], (g) DSAM-Net [50], (h) PATM-Net [58], and (i) proposed HAFF.

summarized in Table II in terms of ten popular metrics. The text set in boldface marks the best accuracy for one dataset. The comparisons substantially indicated that the proposed approach achieved the best accuracies for almost all evaluation metrics. For example, the proposed approach obtained OA = 99.10%, Ka = 0.8325, and AA = 87.73% for Data-A. The resulting accuracies are of the proposed are superior to the ones obtained with Siam-U-net-Conc [37], Siam-U-net-Diff [37], SNU-Net [40], DSAM-Net [50], and PATM-Net [58]. The visual comparisons presented in Figs. 6–8 clearly support the obtained quantitative results. All detection maps obtained using our proposed approach performed with more accurate change detection and fewer missed and false detection pixels.

2) *Comparison Based on the Bitemporal Heterogeneous RSIs (Data-E to -F):* Similar to the comparative experiments based on bitemporal homogenous RSIs, comparisons based on bitemporal heterogeneous RSIs were conducted as well. Table III shows that when the proposed approach was applied on Data-E to -F, the best accuracies could be obtained with it when compared with the results obtained with Siam-U-net-Conc [37], Siam-U-net-Diff [37], SNU-Net [40], DSAM-Net [50], and PATM-Net [58]. The proposed approach obtained an OA = 99.17% for Data-D, which is the best obtained accuracy when compared with that of the other methods. The corresponding visual comparisons for Data-D to -E are presented in Figs. 9–11. Moreover, the visual observation clearly supported the quantitative conclusion in Table III.

Overall, the quantitative and qualitative comparisons in these experiments demonstrated the following outcomes: 1) the proposed approach is feasible to obtain a satisfactory change detection map; 2) it is robust to homogeneous and heterogeneous RSIs for LCCD tasks; and 3) finally, it can achieve a better change detection map when compared with the maps obtained by the cognate methods, namely

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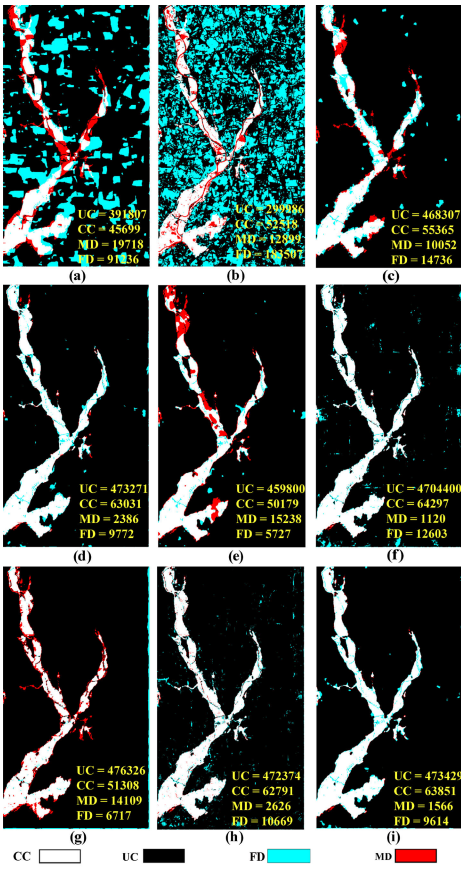


Fig. 10. Detection maps acquired with different methods for Data-E (a) MLS [20], (b) CVA-FCM [21], (c) AGSCC [25], (d) Siam-U-net-Conc [37], (e) Siam-U-net-Diff [37], (f) SNU-Net [40], (g) DSAM-Net [50], (h) PATM-Net [58], and (i) proposed HAFF.

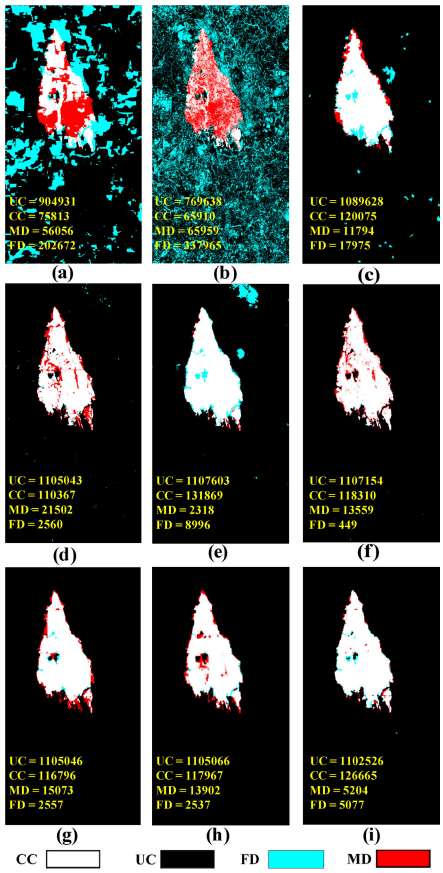


Fig. 11. Detection maps acquired with different methods for Data-F (a) MLS [20], (b) CVA-FCM [21], (c) AGSCC [25], (d) Siam-U-net-Conc [37], (e) Siam-U-net-Diff [37], (f) SNU-Net [40], (g) DSAM-Net [50], (h) PATM-Net [58], and (i) proposed HAFF.

Siam-U-net-Conc [37], Siam-U-net-Diff [37], SNU-Net [40], DSAM-Net [50], and PATM-Net [58].

D. Discussion

In order to give guidance for the practical applications of the proposed approach, the effectiveness and setting of parameters for the compound loss function of the proposed approach are discussed below.

1) Aiming at investigating the performance of the proposed approach in terms of the CMIs [68], the ROC and PR curves of CMIs were generated and compared for each method and dataset in Fig. 12. From the observation in Fig. 12, the observation indicated that our proposed HAFF-based network can achieve better performance on the ROC and PR curves of most datasets, compared with that of other methods. Besides the above visual observation, the quantitative comparisons were summarized in Tables V and VI, and the quantitative observation well supported the conclusion of visual observation. From the comparisons in Tables V and VI, it can see that the proposed HAFF performed better two time out of six datasets, whereas Siam-U-net-Conc [37] and Siam-U-net-Diff [37] performed better four times out of six datasets in terms of AUR, AUP, PRE, and REC. Because the unchanged and changed samples are highly imbalance.
2) *Ablation Experiments to Verify the Effectiveness of the Proposed MFFC Module:* Tables VII and VIII show that the backbone is the classical U-Net, and PAM and MCFF denote the PAM and MCFF) modules, respectively. The comparison in Table VII indicates that when the backbone is coupled with PAM or MFFC, all detection accuracies are improved compared with that using the backbone alone for a homogeneous dataset. Moreover, when the proposed MFFC is coupled with PAM in the U-Net backbone, the best accuracies were achieved in terms of all evaluation metrics. The ablation experiments substantially proved the effectiveness of the proposed MFFC module in improving the detection accuracies for LCCD with both homogeneous and heterogeneous RSIs.
3) *Discussion on the Parameter Setting for the Compound Loss Function of the Proposed Approach:* Table IX shows that although the detection performance of the proposed approach is not sensitive to the setting of $\lambda_1, \lambda_2, \lambda_3$, and λ_4, it can be observed from our experimental investigation of the setting of $\lambda_1, \lambda_2, \lambda_3$, and λ_4, that when $\lambda_1 = \lambda_2 = \lambda_3 = 0.2$, and $\lambda_4 = 0.4$, the proposed approach acquired the best detection accuracies. The reason being that the role of λ_1, λ_2, and λ_3 lies in balancing the background's affection on identifying change areas. In theory, we assumed

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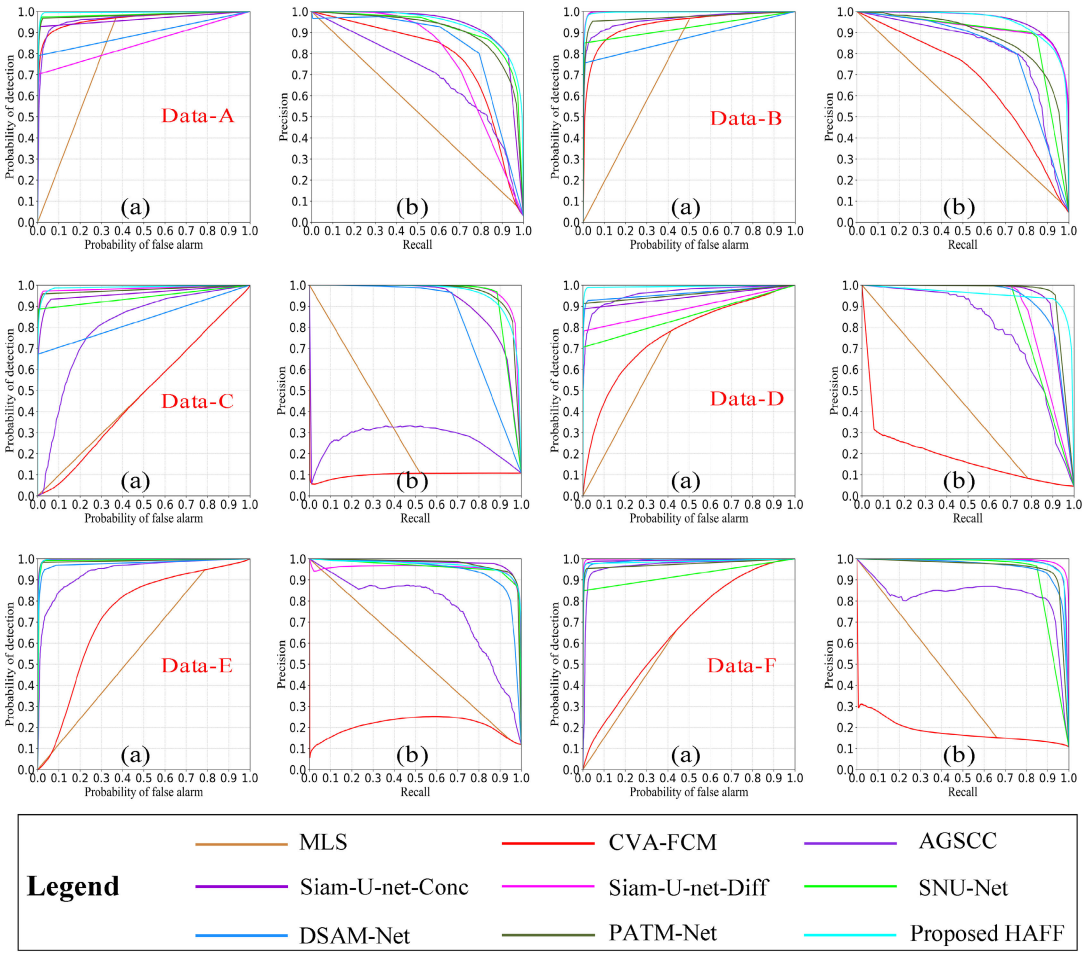


Fig. 12. Comparisons of ROC and PR curves CMIs generated with different methods for each dataset (a) ROC curves and (b) PR curves.

TABLE VII

ABLATION EXPERIMENT COMPARISON BASED ON DATA-B (HOMOGENEOUS RSIS)

Models	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	F1
Backbone	98.22	0.86	94.89	1.06	9.00	1.59	77.59	97.08	87.34	87.86
Backbone + PAM	98.63	0.87	95.12	0.99	8.81	1.37	77.74	97.51	87.63	87.99
Backbone + MFFB	98.74	0.87	95.47	0.86	8.43	1.21	78.43	98.66	88.55	88.12
Backbone + PAM + MFFB	**98.80**	**0.88**	**95.55**	**0.85**	**8.06**	**1.20**	**78.85**	**98.75**	**88.80**	**88.18**

TABLE VIII

ABLATION EXPERIMENT COMPARISON BASED ON DATA-F (HETEROGENEOUS RSIS)

Models	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	F1
Backbone	98.66	0.94	96.84	0.86	4.82	1.37	90.10	96.14	93.12	93.47
Backbone + PAM	98.90	0.95	96.98	0.78	4.54	1.16	91.24	97.66	94.45	94.81
Backbone + MFFC	98.96	0.95	97.25	0.53	3.83	0.91	90.80	99.04	94.92	95.29
Backbone + PAM + MFFC	**99.17**	**0.96**	**97.80**	**0.46**	**3.95**	**0.83**	**92.49**	**99.08**	**95.78**	**96.10**

intuitively that the features from different convolution layers share the same background information. Hence, when they are given equal weight, the total propagation error can be reduced.

4) Besides the above discussions, the detection accuracies were compared when applying the proposed approach with different ratio of training samples. As shown in Fig. 13, it can see that the detection accuracies were

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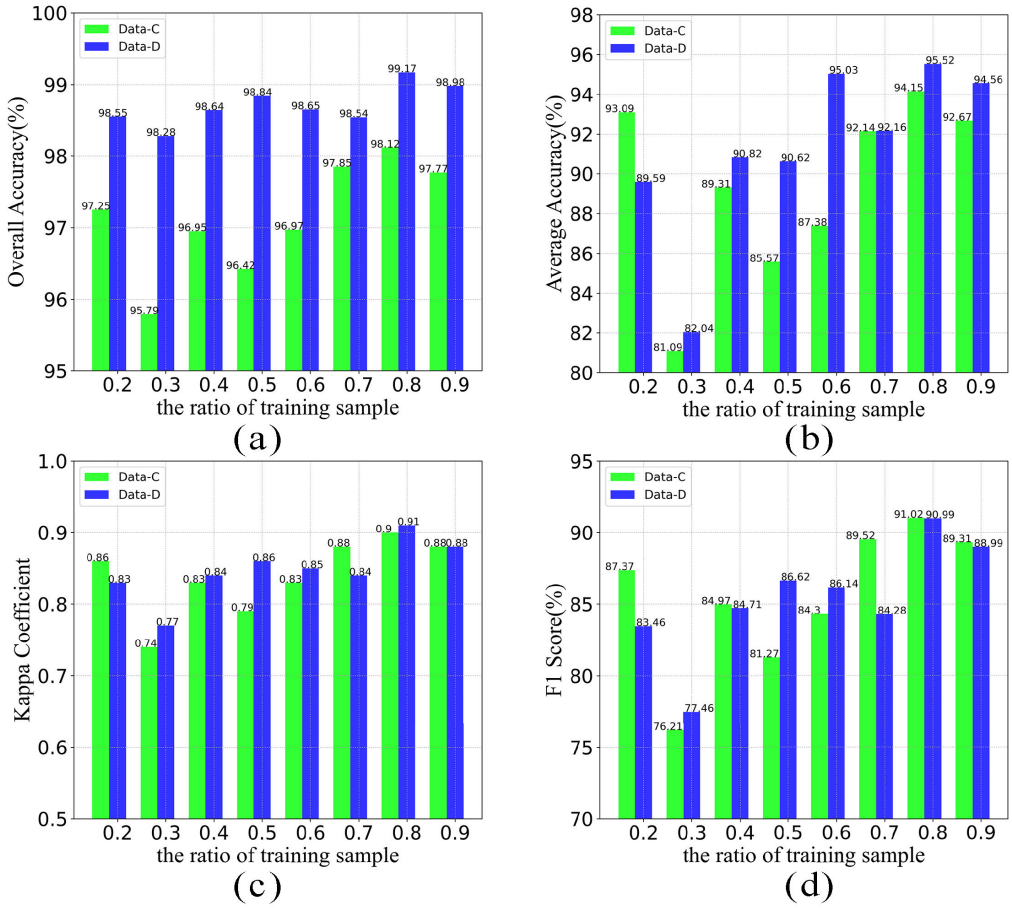


Fig. 13. Accuracy comparisons obtained by applying the proposed approach with different TR, where TR is the ratio between training samples and the total samples. (a) Overall accuracy. (b) Average accuracy. (c) Kappa coefficient. (d) F1 score.

TABLE IX
PARAMETER SETTING FOR THE COMPOUND LOSS FUNCTION OF THE PROPOSED APPROACH FOR DATA-F

$\lambda 1$	$\lambda 2$	$\lambda 3$	$\lambda 4$	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	F1
0.25	0.25	0.25	0.25	99.11	0.95	97.58	0.49	4.06	0.93	92.10	98.94	95.52	95.85
0.10	0.20	0.30	0.40	99.12	**0.97**	97.25	0.50	4.01	1.01	92.35	99.01	95.68	96.01
0.00	0.00	0.00	1.00	98.93	0.95	96.89	0.49	4.80	1.39	91.00	98.12	94.56	94.91
0.15	0.20	0.20	0.45	99.08	0.95	97.72	0.49	4.11	0.88	92.23	98.43	95.33	95.66
0.15	0.15	0.20	0.50	98.75	0.95	96.31	0.58	4.09	1.07	89.17	98.07	93.62	93.96
0.05	0.10	0.10	0.75	99.06	0.95	97.06	0.46	4.99	0.90	91.52	98.86	95.19	95.52
0.20	**0.20**	**0.20**	**0.40**	**99.17**	0.96	**97.80**	**0.46**	**3.95**	**0.83**	**92.49**	**99.08**	**95.78**	**96.10**

improved with the increasement of training samples. However, the detection accuracies not always increase linearly, for example, when the ratio between training samples and the total samples is 0.3.

IV. CONCLUSION

An HAFF-based network for LCCD with RSIs is proposed in this article. In the proposed network, we concentrate on developing an effective feature fusion strategy in deep learning for LCCD. The feasibility and superiority of the proposed network have been verified by comparing it with eight state-of-the-art cognate methods with six pairs of real RSIs. The major advantages of the proposed HAFF are summarized as follows.

1) The proposed network achieves excellent accuracies. The comparative case studies based on six pairs of real RSIs demonstrated that the our approach achieved better performance than eight widely used cognate methods: MLS [20], CVA-FCM [21], AGSCC [25], Siam-U-net-Conc [37], Siam-U-net-Diff [37], SNU-Net [40], DSAM-Net [50], and PATM-Net [58].
2) Coupling the novel MCFF module in the proposed HAFF-based network is effective for improving change detection performance. Moreover, the improvement is

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robust to LCCD with homogeneous and heterogeneous RSIs. The ablation experiment presented here significantly indicates that the proposed MCFF module is effective in improving detection accuracies.

3) A simple but effective neural network is preferred and acceptable for practical applications. The proposed HAFF-based network is an extension of the classical U-Net. The architecture and idea of the proposed HAFF-based network is simple, intuitive, and effective in acquiring a better change detection map than some cognate neural networks. This characteristic results in easy practical application.

Although the proposed HAFF-based network has been verified with six pairs of real RSIs related to landslides, floods, wildfires, and building construction change events in the real world, the study areas are evidently small. In our future research, a dataset for describing change detection events covering large geographic areas will be collected to further investigate the practicality and robustness of the proposed approach. In addition, owing to the advantages of graph representation, graph convolutions will be used to reconstruct an image block in our proposed approach to further enhance the detection performance.

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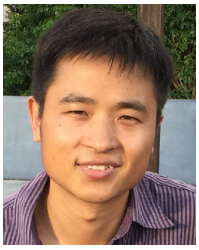
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